

Spring 2021 Real Analysis II — Homework 4

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1. Let X and Y be Banach spaces. Show that $T \in L(X, Y)$ if and only if $T : X \rightarrow Y$ is linear and $T(U) \subset Y$ is bounded whenever $U \subset X$ is bounded.

Proof. ($\Rightarrow$) Obvious.

($\Leftarrow$) Since $\bar{B}(0; 1) = \{x \in X : \|x\| \leq 1\}$ is bounded in X , we know $T(\bar{B}(0; 1))$ is bounded, i.e., there exists $M > 0$ such that $\|Tx\| \leq M$ for all $x \in \bar{B}(0; 1)$. Hence for any nonzero $x \in X$, we have $x/\|x\| \in \bar{B}(0; 1)$ and

$$\left\| T \frac{x}{\|x\|} \right\| = \frac{1}{\|x\|} \|Tx\| \leq M,$$

which implies that $\|Tx\| \leq M\|x\|$ for all $x \in X$. Hence $T \in L(X, Y)$. $\square$

2. Let X and Y be Banach spaces and $T \in L(X, Y)$. Show that

- (a) $\|T\| = \sup_{\|x\| \leq 1} \|Tx\|$.
- (b) $\|T\| = \sup_{\|x\| < 1} \|Tx\|$.
- (c) $\|T\| = \inf \{M \in \mathbb{R} : \frac{\|Tx\|}{\|x\|} \leq M, \forall x \in X \setminus \{0\}\}$.

Proof. (a) It is clear that $\|T\| \leq \sup_{\|x\| \leq 1} \|Tx\|$. Now we need to show $\|T\| \geq \sup_{\|x\| \leq 1} \|Tx\|$. Without loss of generality, we assume $T \neq 0$. Let $\{x_k\} \subset \bar{B}(0; 1)$ be nonzero such that $\|Tx_k\| \rightarrow \sup_{\|x\| \leq 1} \|Tx\|$. Then we have

$$\|T\| \geq \frac{\|Tx_k\|}{\|x_k\|} \geq \|Tx_k\| \rightarrow \sup_{\|x\| \leq 1} \|Tx\|$$

as $k \rightarrow \infty$. Hence we have $\|T\| \geq \sup_{\|x\| \leq 1} \|Tx\|$.

- (b) Similar as above, we can show $\|T\| \geq \sup_{\|x\| > 1} \|Tx\|$. Now let $x_k \in \partial B(0; 1)$ such that $\|Tx_k\| \rightarrow \|T\|$ as $k \rightarrow \infty$. Let $y_k \in B(0; 1)$ such that $\|y_k - x_k\| \rightarrow 0$ as $k \rightarrow \infty$. Then we have

$$\|T\| = \lim_{k \rightarrow \infty} \|Tx_k\| \leq \lim_{k \rightarrow \infty} (\|Ty_k\| + \|T(x_k - y_k)\|) = \lim_{k \rightarrow \infty} \|Ty_k\| \leq \sup_{\|x\| > 1} \|Tx\|.$$

- (c) Denote $S := \{M \in \mathbb{R} : \frac{\|Tx\|}{\|x\|} \leq M, \forall x \in X \setminus \{0\}\}$ and $m := \inf_{M \in S} M$. Then we need to show $\|T\| = m$. It is obvious that $\|T\| \leq m$ since $\|T\| \leq M$ for all $M \in S$. On the other hand, $\|T\| \in S$ so $\|T\| \geq m$. $\square$

3. Let X be a Banach space and $f \in X^*$. For any $x_0 \in X$ and $\delta > 0$, show that

- (a) $\sup_{x \in B(x_0; \delta)} f(x) = f(x_0) + \delta \|f\|$.

(b) $\inf_{x \in B(x_0; \delta)} f(x) = f(x_0) - \delta \|f\|.$

Proof. We have

$$\sup_{x \in B(x_0; \delta)} f(x_0) = \sup_{y \in B(0; 1)} f(x_0 + \delta y) = f(x_0) + \delta \sup_{y \in B(0; 1)} f(y) = f(x_0) + \delta \|f\|.$$

Similarly, we have

$$\inf_{x \in B(x_0; \delta)} f(x_0) = \inf_{y \in B(0; 1)} f(x_0 + \delta y) = f(x_0) - \delta \inf_{y \in B(0; 1)} -f(y) = f(x_0) - \delta \| -f \| = f(x_0) - \delta \|f\|.$$

□

4. Let X be a Banach space and $f \in X^*$ be nonzero. Show that there exists no open ball $B(x_0; \delta)$ such that $f(x_0)$ is an extreme value of f on $B(x_0; \delta)$.

Proof. Assume not, then by the previous problem we know $\|f\| = 0$ which contradicts to f being nonzero. □

5. Let $X = L^1([a, b])$ and $Y = C([a, b])$. Define $T : X \rightarrow Y$ by

$$(Tf)(x) = \int_a^b f(t) dt, \quad \forall f \in X.$$

- (a) Find the norm of $T \in L(X, Y)$.
(b) Find the norm of $T \in L(X, X) =: L(X)$.

Proof. (a) We have for any $f \in X$ that

$$\|Tf\|_Y = \max_{a \leq x \leq b} \left| \int_a^x f(t) dt \right| \leq \max_{a \leq x \leq b} \int_a^x |f(t)| dt \leq \int_a^b |f(t)| dt = \|f\|_X.$$

Hence $\|T\| \leq 1$. On the other hand, let $f(t) = \frac{1}{b-a}$ for all $t \in [a, b]$, then

$$\|Tf\|_Y = \max_{a \leq x \leq b} \left| \int_a^x \frac{1}{b-a} dt \right| = 1 = \|f\|_X.$$

Hence $\|T\| = 1$.

- (b) We have for any $f \in X$ that

$$\|Tf\|_X = \int_a^b |(Tf)(x)| dx = \int_a^b \int_a^x |f(t)| dt dx \leq \int_a^b \int_a^b |f(t)| dt dx = (b-a) \|f\|_X.$$

Hence $\|T\| \leq b-a$. Let $f_k(t) = ke^{-k(t-a)}$ for any $t \in [a, b]$, then

$$\|f_k\|_X = \int_a^b ke^{-k(t-a)} dt = 1 - e^{-k(b-a)}.$$

On the other hand, we have

$$\|Tf_k\|_X = \int_a^b \int_a^x ke^{-k(t-a)} dt dx = \int_a^b (1 - e^{-k(x-a)}) dx = (b-a) + \frac{1}{k}(e^{-k(b-a)} - 1).$$

Hence there is $\|Tf_k\|_X / \|f_k\|_X \rightarrow b-a$ as $k \rightarrow \infty$. Therefore $\|T\| = b-a$.

□

6. Let X be a Banach space and $f \in X^*$. Show that for any $\epsilon > 0$ there exists $x_0 \in X$ such that $f(x_0) = \|f\|$ and $\|x_0\| \leq 1 + \epsilon$.

Proof. By the definition of functional norm, we know for any $\epsilon > 0$, there exists $x \in X$ such that $\|x\| = 1$ and $f(x) = \|f\|(1 - \delta)$ where $\delta = \frac{\epsilon}{1+\epsilon} \in (0, 1)$. Let $x_0 = \frac{x}{1-\delta}$, then it is easy to verify that $f(x_0) = \|f\|$ and $\|x_0\| = \frac{1}{1-\delta} = 1 + \epsilon$. $\square$

7. Let X and Y be Banach spaces and $T : X \rightarrow Y$ be linear. Denote $Z(T) := \{x \in X : Tx = 0\}$.

- (a) Suppose $T \in L(X, Y)$. Show that $Z(T)$ is a closed linear subspace of X .
(b) Suppose $Y = \mathbb{R}$. Show that $T \in X^*$ if and only if $Z(T)$ is a closed linear subspace of X .

Proof. (a) It is clear that $Z(T)$ is a linear subspace of X . Let $x_k \in Z(T)$ such that $x_k \rightarrow x$. As $T \in L(X, Y)$, we know $T(x_k) \rightarrow T(x)$ which implies that $T(x) = 0$. Hence $x \in Z(T)$. Therefore $Z(T)$ is closed.

- (b) Necessity is shown as above. For sufficiency, suppose T is not bounded, then there exists $x_k \in X$ such that $\|x_k\| = 1$ but $f(x_k) \geq k$ for all $k \in \mathbb{N}$. Denote $y_k = \frac{x_k}{f(x_k)} - \frac{x_1}{f(x_1)}$, we have $f(y_k) = 0$ for all $k \in \mathbb{N}$ and

$$\left\| y - \frac{x_1}{f(x_1)} \right\| = \frac{\|x_k\|}{f(x_k)} \leq \frac{1}{k} \rightarrow 0.$$

Hence $y_k \in Z(T)$ and $y_k \rightarrow x_1/f(x_1)$, but $x_1/f(x_1) \notin Z(T)$, which contradicts to $Z(T)$ being closed. Hence T is bounded and therefore $T \in X^*$. $\square$

8. Let X be a Hilbert space and $f \in X^*$. Suppose $C \subset X$ is a closed convex set and $g : C \rightarrow \mathbb{R}$ is defined by $g(x) := \frac{1}{2}\|x\|^2 - f(x)$ for any $x \in C$. Show that

- (a) There exists a unique $y \in X$ such that $g(x) = \frac{1}{2}\|x - y\|^2 - \frac{1}{2}\|y\|^2$ for any $x \in C$.
(b) There exists a unique $z \in C$ such that $g(z) = \inf_{x \in C} g(x)$.

Proof. (a) By Riesz representation theorem, there exists a unique $y \in X$ such that $f(x) = \langle x, y \rangle$ for all $x \in X$. Hence

$$g(x) = \frac{1}{2}\|x\|^2 - f(x) = \frac{1}{2}\|x\|^2 - \langle x, y \rangle = \frac{1}{2}\|x - y\|^2 - \frac{1}{2}\|y\|^2.$$

- (b) We know that there exists a unique $z \in C$ as the projection of y onto C , such that $\|z - y\| = \inf_{x \in C} \|x - y\|$. Therefore

$$g(z) = \frac{1}{2}\|z - y\|^2 - \frac{1}{2}\|y\|^2 = \inf_{x \in C} \left(\frac{1}{2}\|x - y\|^2 - \frac{1}{2}\|y\|^2 \right) = \inf_{x \in C} g(x).$$

$\square$

9. Let S be a set and $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space, where the elements of X are real-valued functions defined on S . For any $x \in S$, define $J_x \in X^*$ by $J_x(f) := f(x)$. Show that there exists $K : S \times S \rightarrow \mathbb{R}$ such that

- (a) For any $y \in S$, $K(\cdot, y) \in X$ is a real-valued function defined on S .

(b) For any $f \in X$ and $y \in S$, there is $f(y) = \langle f, K(\cdot, y) \rangle$.

Such function K is called a *reproducing kernel of the Hilbert space X* , and X is called a *reproducing kernel Hilbert space (RKHS)*.

Proof. By Riesz representation theorem, for any $y \in S$, there exists a unique $K_y \in X$ such that $J_y(f) = \langle f, K_y \rangle = f(y)$. Define $K : X \times X \rightarrow Y$ such that $K(x, y) := K_y(x)$. Then obviously K satisfies both (a) and (b). $\square$

10. Let S and X be defined as above, and $\{e_k : k \in \mathbb{N}\}$ be a basis of X . Suppose K is a reproducing kernel. Show that $K(x, y) = \sum_{k=1}^{\infty} e_k(x)e_k(y)$ for all $x, y \in S$.

Proof. By the previous problem, we know for any $x, y \in X$, there is $K(x, y) = K_y(x) = \sum_{k=1}^{\infty} a_k e_k(x)$ where

$$a_k = \langle e_k, K_y \rangle = \langle e_k(\cdot), K(\cdot, y) \rangle = e_k(y)$$

due to the property of K . Hence $K(x, y) = \sum_{k=1}^{\infty} e_k(x)e_k(y)$. $\square$